

Data Storage & Saved Sheets

How the app remembers your work on the device

⚠ FOR UNDERSTANDING PURPOSES ONLY

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Flutter / Dart

One database

5-slot saves

Reports log

Safe upgrades

1. The Idea

Almost every working screen in DrillTick — a kill sheet, a permit, a checklist, a report — can be saved and reopened later. Behind all of them sits **one local database on the device**. There are no accounts and no servers; saved work stays on the phone.

2. What Is Kept

Data	How it is stored
Rigging equipment records	A table of equipment items, plus user-added categories and each item's inspection history (every inspection is kept, newest first)
Crane planning	Saved slots for crane load charts, lift plans and tandem lifts
Kill sheets	Five numbered slots for the Surface sheet and five for the Subsea sheet
Permit to Work	Three numbered slots for each permit type
Checklists	Five slots for the SIMOPS Matrix, with a ready pattern for adding more checklists
HSE reporting	Three slots for the Monthly Report, plus permanent logs of every submitted incident report, near-miss report and observation

3. The Slot Idea

A **slot** is a numbered save position, like a save-game slot. Each slot stores a short label you choose, the time it was saved, and the whole form's contents.

1. **Save.** Pick a slot, give it a name and press save. If the slot already holds something, it is replaced.

2. **Open.** A picker lists every slot with its name and time; empty slots are shown as empty.
3. **Delete.** A slot can be cleared on its own.

Why the picker is quick

The label and the time are kept separately from the bulky form contents. The picker only reads the short labels, and the full form is loaded only when a slot is actually opened.

4. Append-Only Logs

Submitted incident reports, near-miss reports and observations are different from slots. They are **logs**: each submission is added as a new record with a timestamp and is never overwritten. The three kinds of observation (safe, unsafe act and unsafe condition) share one log, tagged by type, so the Safety Dashboard can count and compare them.

5. Growing the Database Safely

The database has gone through **seven versions** as features were added: equipment first, then crane slots, kill sheets, inspections, permits, reporting and finally checklists.

- Each upgrade only **adds** tables; existing data is never rebuilt.
- Tables are created only “if they don’t already exist”, so an upgrade can safely run twice.
- Newer screens also re-check that their table exists every time they save, which protects phones that skipped an earlier upgrade.
- A one-time **migration** moves any permit saves made by the older file-based system into the database, then removes the old files.

6. Why a Database Instead of Files

The kill sheets and permits originally saved each slot as a small file. They were moved into the database so saves survive reinstalls, cache clearing and device backup and restore. The two SCR Measure screens still use the simpler on-device file approach, with three slots each.